TUM School of Computation, Information and Technology
Technical University of Munich

Semester Thesis

English Title

German Title

Supervisor Prof. Dr.-Ing. habil. Alois C. Knoll

Advisor Zhang Chi, M.Sc.

Author Carsten Oertel

Date April 15, 2025 in Munich

Disclaimer

I confirm that this Semester Thesis is my own work and I have documented all sources and material used.

Munich, April 15, 2025

(Carsten Oertel)

Abstract

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Kurzfassung

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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Chapter 1

About TUMlatex

1.1 Introduction

This TUMlatex project represents a cooperation of several chairs of the Technical University of Munich to provide a compact and easy-to-use LaTeX package for both, employees and students.

If you are interested in using this package, please read Section 1.2.

If you are an advanced LaTeX user and want to contribute to this project, please read Section 1.3.

If you are responsible for the LaTeX templates of your chair and you are interested in joining this collaboration, please contact the current maintainer Michael Kreutz (m.kreutz@tum.de).

1.2 Quick-Start-Guide for Beginners

This project aims to provide an easy-to-use solution for unexperienced or even first-time LaTeX users. As a first step, please read the installation instructions of Section 1.2.1.

1.2.1 Installation

If you are using TUMlatex for the first time, please pass through the following steps carefully.

Setup your LaTeX Environment

First of all you have to install a LaTeX environment on your PC. You may skip this step, if you are using a PC of the university with a pre-installed LaTeX environment (e.g. TUM-PC).

- For **Windows** users: Please install the newest version of MiKTeX (see below for full installation).
- For **Mac** users: Please install the newest version of MacTeX (see below for full installation).
- For **Linux** (Ubuntu, TeXLive) - Option A (outdated version - not recommended): Install the default packages texlive-full and biber via apt-get shipped with your Ubuntu distribution.

- For **Linux** (Ubuntu, TeXLive) - Option B (**newest version - recommended**): Download the TUMlatex package of your chair (see Section 1.2.1), unzip it and run the automatic installer script `UbuntuInstallTexliveNewest.sh` as root (`sudo ./UbuntuInstallTexliveNewest.sh`) located in XXXTools (replace XXX by your chair abbreviation).

Note: Without any warranty you may use a different LaTeX environment.

Important: If the amount of free memory on your PC permits, be sure to install all packages. Otherwise it is likely, that you are missing some necessary packages and have to install them on your own.

- For **Windows** users: Install all packages via the MiKTeX Package Manager.
- For **Mac** users: Install the Full MacTeX package.
- For **Linux** users: Both options (A and B) mentioned above will install a full installation of TeXLive.

Optional: Update your LaTeX Environment

In some cases this is necessary in order to avoid compilation errors due to outdated standard packages. Again you may skip this step if this is not your private PC.

- For Windows users: Please follow this guide.
- For Mac users: You can use the graphical updating tool TeX Live Utility to update your packages.
- For Linux users: If you chose option B (see Section 1.2.1) open a terminal and update the TeXLive package manager and all packages with the command `sudo tlmgr update -self -all -reinstall-forcibly-removed`

Setup your LaTeX Editor

You can use any editor you like. However, the open-source and cross-platform LaTeX editor TeXstudio is recommended. In this case you can use the **auto-completion feature** by copying the provided syntax file (`.cwl-file`, shipped within the `.zip` file, see Section 1.2.1) to the folder

- `%APPDATA%\Roaming\texstudio\completion\user` for Windows users or equivalently
- `~/.config/texstudio/` for Linux or Mac users.

Furthermore you may enable the shell-escape option of pdflatex (optional, but neccessary for tikz-externalize to work):

1. Navigate to *Options* → *Configure TeXstudio* → *Commands*
2. Change *PdfLaTeX* to `pdflatex -synctex=1 -shell-escape -interaction=nonstopmode %.tex`

You also should change the default bibliography tool from *bibtex* to ***biber***:

1. Navigate to *Options* → *Configure TeXstudio* → *Build*
2. Change *Default Bibliography Tool* to `txs:///biber`

Note: You can also use bibtex as bibliography backend with the TUMlatex packages. However, you have to pass `optBibtex` *instead of* `optBiber` *to the package. Furthermore some commands related to the bibliography are different (ask Google for help).*

Get the latest Release of the TUMlatex Package

If you are reading this document you probably already have downloaded the TUMlatex package. However, it is recommended to check if you are using the latest version. In the following table you find the chair-related TUMlatex packages:

Dept.	Chair	TUMlatex Package
ED	Aerodynamics and Fluid Mechanics	AERlatex.zip
ED	Applied Mechanics	AMlatex.zip
ED	Automotive Technology	FTMlatex
ED	Machine Tools and Industrial Management	IWBlatex.zip
ED	Central Teaching Unit	ZLlatex.zip
CIT	I6 - Robotics, AI & Embedded Systems	AIRlatex.zip

The packages include (replace "`XXX`" by your chair abbreviation)

- the main class file `XXXlatex.cls`,
- the main package file `XXXlatex.sty` (This is an alternative to using the main class file. This is helpful, if you write a paper and have to use the baseclass provided by the publisher),
- the syntax file `XXXlatex.cwl` for auto-completion in TeXstudio,
- a license file `XXXlatex_LICENSE.txt` containing licensing information,
- a folder `XXXDocumentation` containing the package documentation (therein precompiled as `main.pdf`),
- a folder `XXXTemplates` containing a selection of useful templates which you can copy and directly use.
- a folder `XXXTools` containing a selection of useful tools (e.g. TeXLive setup script).

1.2.2 Create your own Document

The **recommended way** to create a new document is to

1. select an appropriate template in the template folder `XXXTemplates`,
2. **copy the whole template folder** (e.g. `XXXDocument`) wherever you want,
3. open the `main.tex` file in the template folder with your favorite LaTeX-editor,
4. initially compile it (using *pdflatex* and optionally *biber/bibtex*) and check for errors,

5. make your changes and fill it with content and
6. compile it (using *pdflatex* and optionally *biber/bibtex*).

Note: As an alternative way it is possible to use the main class (`.cls`-file) or the main package (`.sty`-file) as a shared resource by copying it to your local `texmf` folder. However, in this case you may not be able to compile old documents, which have been created with a different version of this package. Thus it is ***strongly recommended*** *to make a copy of the main class file (`XXXlatex.cls`) for each of your documents!*

1.2.3 Documentation

The documentation of the packages is included in the folder `XXXDocumentation` (precompiled as file `main.pdf`). Feel free to have a look at the source code of the documentation, in order to learn how to do stuff. Be aware that most of the source code is automatically generated.

1.2.4 Bugs, Errors and Suggestions

If you recognize any bug/error or you have a suggestion to make this package (even) better, please feel free to open an Issue.

Important: Assign the corresponding maintainer of your chair to the issue (see LRZ GitLab). Expect no support for unassigned issues - they will be deleted immediately!

Please use the default **Labels** to categorize your issue. Please also provide additional information like your

- operating system (+ version),
- LaTeX environment (+ version) and
- LaTeX editor and (+ version).

1.3 Guide for Developers/Contributors

For information on how to contribute to this project please have a look at the Contribution Guide.

1.4 Support

For support, please directly contact the responsible package maintainer of you chair (see "maintainer" in Section 1.2.4).

Important: If your are not associated with a chair or your chair is not part of the TUMlatex project, please read following note: *We are happy that you use TUMlatex and that you appreciate our effort in making a good LaTeX template for TUM members. However, please understand that this project is maintained by various PhD students, who contribute to the template mainly in their spare time (actually we do not get paid for this - its our personal commitment). Thus, we limit our "support" to employees and students of our associated chairs. Certainly, we are happy for any bug report, however, we can not assist everybody in using this template neither can we respond to individual feature requests. We hope you can understand this.*

1.5 License Notes

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This work may be distributed and/or modified under the conditions of the LaTeX Project Public License, either version 1.3 of this license or (at your option) any later version. The latest version of this license is in http://www.latex-project.org/lppl.txt and version 1.3 or later is part of all distributions of LaTeX version 2005/12/01 or later.

This work has the LPPL maintenance status 'maintained'. The current maintainer of this work is Michael Kreutz (m.kreutz@tum.de).

This work consists of all files of the GitLab repository https://gitlab.lrz.de/AM/TUMlatex.

Chapter 2

I6 Submission Guidelines

This chapter gives an overview about the expected format of a thesis at the Chair of Robotics, Artificial Intelligence and Real-time systems. It mostly focuses on the format on your written thesis. For a general overview of the submission guideline please refer to the I6 Wiki.

When writing your thesis please use the `AIRStudentThesis` template. It should be in line with the IN TUM thesis guidelines (last checked: 08/02/2020). However, please always reassure that this is still the case. In other subjects, please make sure that the format meets your individual requirements. When handing it in, please include an additional copy for your advisor. Moreover make sure your that thesis fulfills the following requirements.

2.1 Citations

We recommend that you use `alphadin` citation style when working with bibtex, or `alpha` when using biblatex/biber. Your bibliography should look like this:

[BM92] Besl, P. J. and McKay, N. D. "Method for registration of 3-D shapes". In: Sensor fusion IV: control paradigms and data structures. Vol. 1611. International Society for Optics and Photonics. 1992, pp. 586–606.

[CM92] Chen, Y. and Medioni, G. "Object modelling by registration of multiple range images". In: Image and vision computing 10.3 (1992), pp. 145–155.

When quoting an author word-by-word please include a page number in the reference.

2.2 Figures

Please use vector graphics when including figures or diagrams in your thesis. If you create them with PGF/TIKZ or Inkscape it also possible to add texts directly from Latex to ensure a consistent typesetting. There is an unwritten rule, saying figures not referenced in your text are unnecessary.

When citing a figure include the reference in the figure's label (including the page number).

2.3 Math Notation

When using formulas or pseudocode use the following notation style:

Type	Example	Code (Math Mode)	Comment
General items			
Scalars	a	`a`	lower case, italic
Functions	$\sin(x)$	`\textrmsin(x)`	regular text
Units	$42\,\textrm{Hz}$	`42\,\textrmHz`	half-space between number and unit; unit in regular text
Angles	α	`\alpha`	Greek letters
Absolute value	$\mid a \mid$	`\mid a \mid`	
Modulo	mod	`\textrm{mod}`	
Vectors and matrices			
Vectors	$\boldsymbol{v}$	`\boldsymbol{v}`	lower case, italic, bold
Matrices	$\boldsymbol{M}$	`\boldsymbol{M}`	upper case, italic, bold
Variable indices	$\boldsymbol{x}_i$	`\boldsymbol{x}_i`	
Static indices	$\boldsymbol{x}_{\textrm{min}}$	`\boldsymbol{x}_\textrm{min}`	
Unit vectors	$\textbf{e}_x$	`\textbf{e}_x`	
Identity matrices	$\textbf{Id}_n$	`\textbf{Id}_n`	
Transpose	$\boldsymbol{x}^{\textrm{T}}$	`\boldsymbol{x}^\textrm{T}`	T in superscript and regular font
Vector product (cross product)	$\times$	`\times`	
Dot product	$\cdot$	`\cdot`	
Sets and sequences			
Sets	$A = \{1,2,3\}$	`A = \{ 1, 2, 3 \}`	upper case, italic
Sequence	$\mathcal{A} = \langle 1,2,3 \rangle$	`\mathcal{A} = \langle 1,2,3 \rangle`	
Set without	$A \backslash \{e\}$	`A \backslash \{ e \}`	
Unification	$A \cup \{e\}$	`A \cup \{ e \}`	

Always use single letters for variable names.

2.4 Length

Stundents often ask for a required number of pages. However, there are no fixed page limits or requirements for bachelor's or master's theses. As a point of reference, most bachelor's have around 40 to 60 pages and while master's thesis usually reach 60 to 80 pages (including abstract, table-of-contents, lists of figures, lists of tables and bibliography). If your thesis is going to be longer than 100 pages, your advisor and/or supervisor will most likely not read all of it. Detailed information can be found in your module handbook (see campus.tum.de).

2.5 Cover

The main directory of the `AIRStudentThesis` template contains two `*.tex`: `main.tex` and `cover.tex`. Start by using `main.tex` to write your thesis. Once you have completed your

work and you know how large the spine of the book will be you can use `cover.tex` to generate a matching cover for your printed copies.

2.6 CD or DVD

Note: A CD or DVD is not required for the official copy you submit to the faculty or your program coordinator. If in doubt check your program's website!

For your advisor's/ supervisor's copy of your thesis please add a CD or DVD on the last inner page of your thesis including:

1. A digital copy of your thesis (including its Latex sources)
2. The code you programmed
3. The dataset(s) you used for your evaluation results

As a rule of thumb, the disk should include everything required to reproduce your results.

Chapter 3

Examples

This chapter shows a number of examples that might be useful when you have not worked with LaTex or TUMlatex before.

Testing references from literature 1: [**ChiZhang2026**] from literature [**chen1992object**]

3.1 Figures

This paragraph shows the differences of pixel based images and vector graphics. As you can see in figure fig. 3.1 the `*.png` image becomes quite blurry when scaled to large, while the other, vector based images look flawless at any zoom rate.

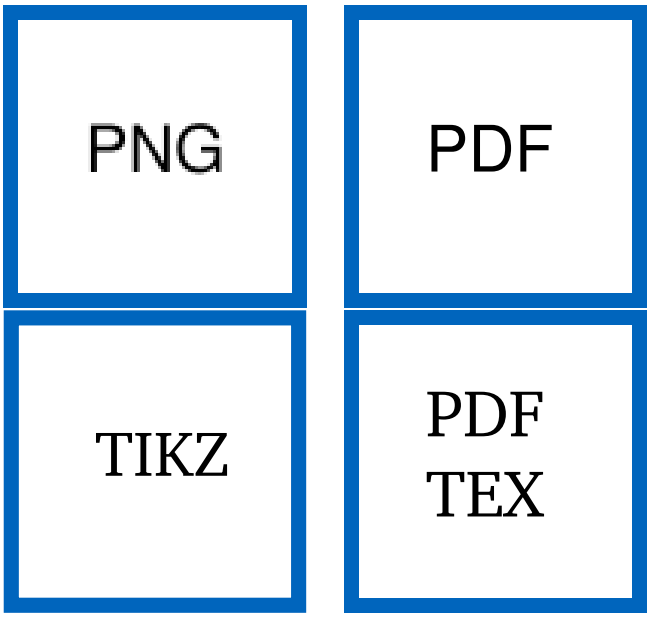

Figure 3.1: Image description.

3.2 Tables

Tables can be referred to in the same way as figures. The four letters `[htbp]` after the `\begin{table}` statement indicate the placement of the table environment as listed in ta-

ble 3.1, where the first letter is the preferred position, the second one the next best alternative and so on.

Letter	Placement
`h`	here
`t`	top of the page
`b`	bottom of the page
`p`	standalone page

Table 3.1: Placement options for floating environments.

We recommend to use the `\toprule`, `\midrule` and `\bottomrule` commands from the `booktabs` package to generate nice looking tables.

3.3 Citations

This sentence is referring to a single source [**besl1992method**]. Sentence number two has a single source with a page number [**chen1992object**] The statement in this last sentence is even supported by multiple sources [**choset2005principles**; **thrun2005probabilistic**].

3.4 Equations

Besides of LaTeX general math mode, that allows you to put some formulas like $a^2 + b^2 = c^2$ directly in your paragraph, we can also use an enumerated, standalone environment for more complex statements, such as Equation 3.1:

$$E = mc^2 \tag{3.1}$$

3.5 Algorithms

When writing your thesis in the field of computer science you might need to include algorithms in your thesis. The `algorithm` and `algorithmicx` offer a floating environment for pseudocode as shown in Algorithm 1.

Algorithm 1 A possible implementation of the bubble sort algorithm

```
Require Array A
n → size(A)
Repeat
    s → false
    for (i → 0; i < n − 1; i → i + 1)
        if A_i > A_{i+1}
            swap(A_i, A_{i+1})
            s → true
While s
```

3.6 Plots

Plots might appear to be just another kind of images, however, *tikz* allows you to define them right in your LaTeX code, as done in figures 3.2 and 3.3.

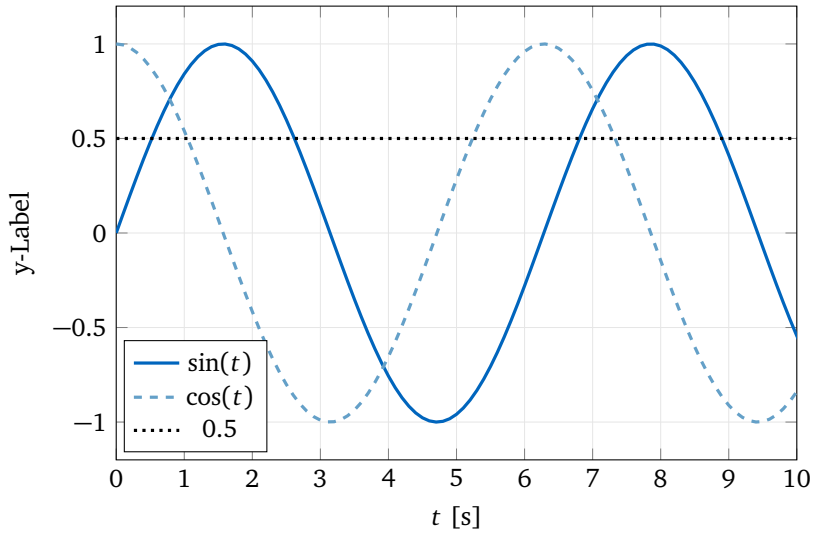

Figure 3.2: Title of the 2D plot (using pgfplots).

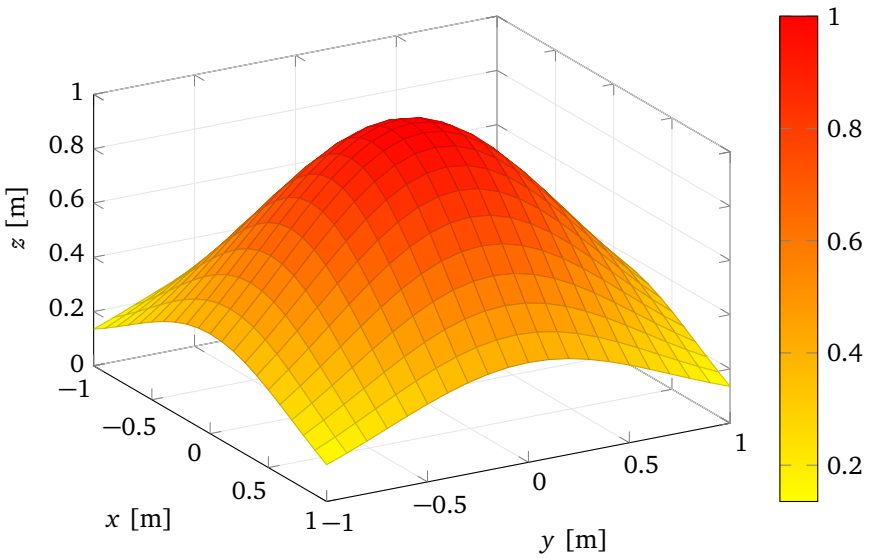

Figure 3.3: Title of the 3D plot (using pgfplots).

3.7 Sections

Of course it is also possible to define deeper levels of sections. Let's have a look at some `subsections` and `subsubsections`. However, when writing you thesis, try to avoid deeply nested structures.

3.7.1 Subsection 1

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Subsubsection 1

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Subsubsection 2

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

3.7.2 Subsection 2

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

3.7.3 Subsection 3

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Appendix A

Appendix 1

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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